

Multimedia data

Hoa Dinh Nguyen, PhD

hoand@ptit.edu.vn

Types of Multimedia data

- Image data
- Audio data
- Text data
- Video data

Multimedia Data — Core Characteristics

- Large volume: images/video/audio dominate storage vs text metadata
- Temporal continuity: audio/video are time-series; images are static 2D fields
- Composite structure: video = image sequence + audio + text (subtitles)
- Semantic richness: human meaning exceeds raw pixel/sample values
- Format heterogeneity: JPEG, PNG, MP3, H.264, HTML, PDF
- Drives MMDB design: LOB storage, indexes on features, not only keywords

Image data

- Images are presented 2D matrices
- Each component of a matrix is presented by
 - Coordinates (i, j)
 - Color values: depends on the color system

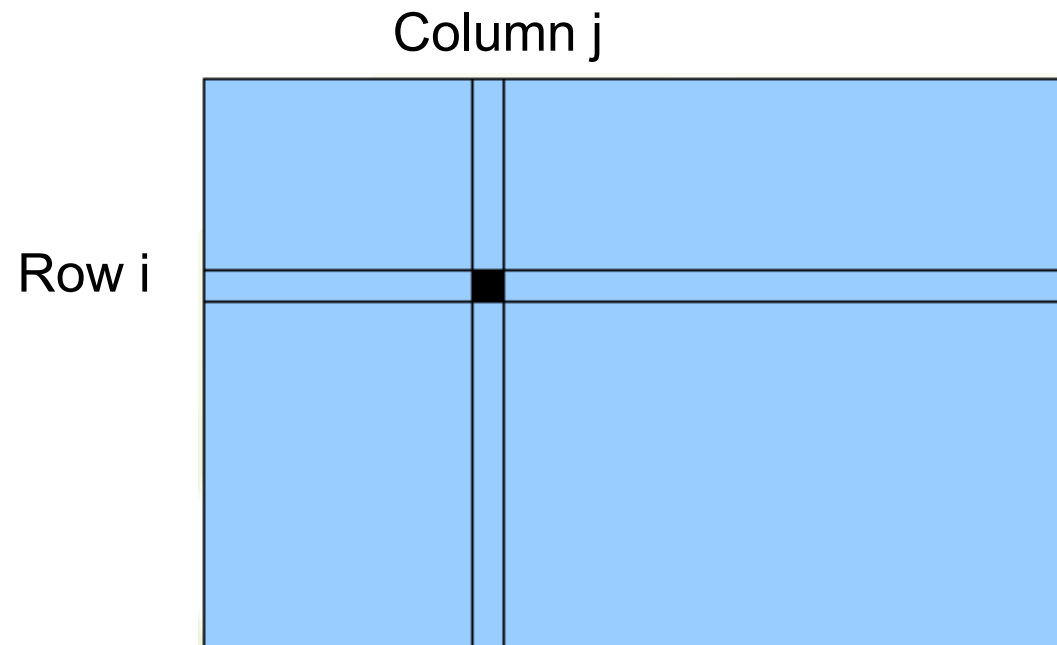
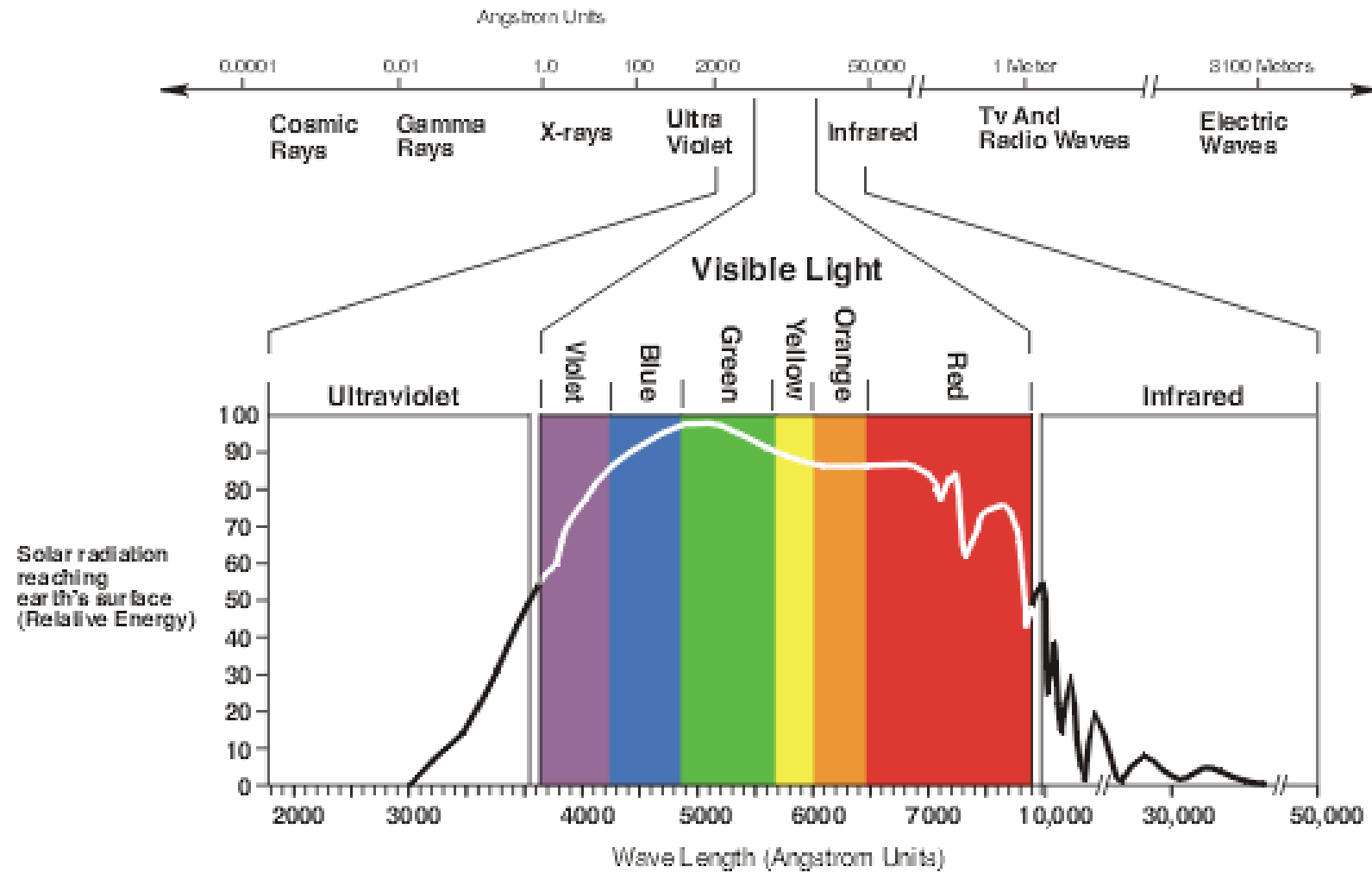


Image Data as a 2D Digital Matrix

- Image = 2D array of pixels; each pixel at coordinates (row i , column j) (slide 3)
- Grayscale: one intensity value per pixel (8 bits $\rightarrow$ 256 levels)
- Color: 3+ values per pixel (R,G,B) or indexed palette
- Resolution: width $\times$ height pixels; bit depth $\times$ channels = bits per pixel
- Example: 1920 $\times$ 1080 RGB 8-bit $\rightarrow$ 1920 $\times$ 1080 $\times$ 3 $\approx$ 6.2 MB uncompressed
- Foundation for Lecture 11 CBIR (color histograms, shape features)

Electromagnetic Spectrum



Fundamentals of colors

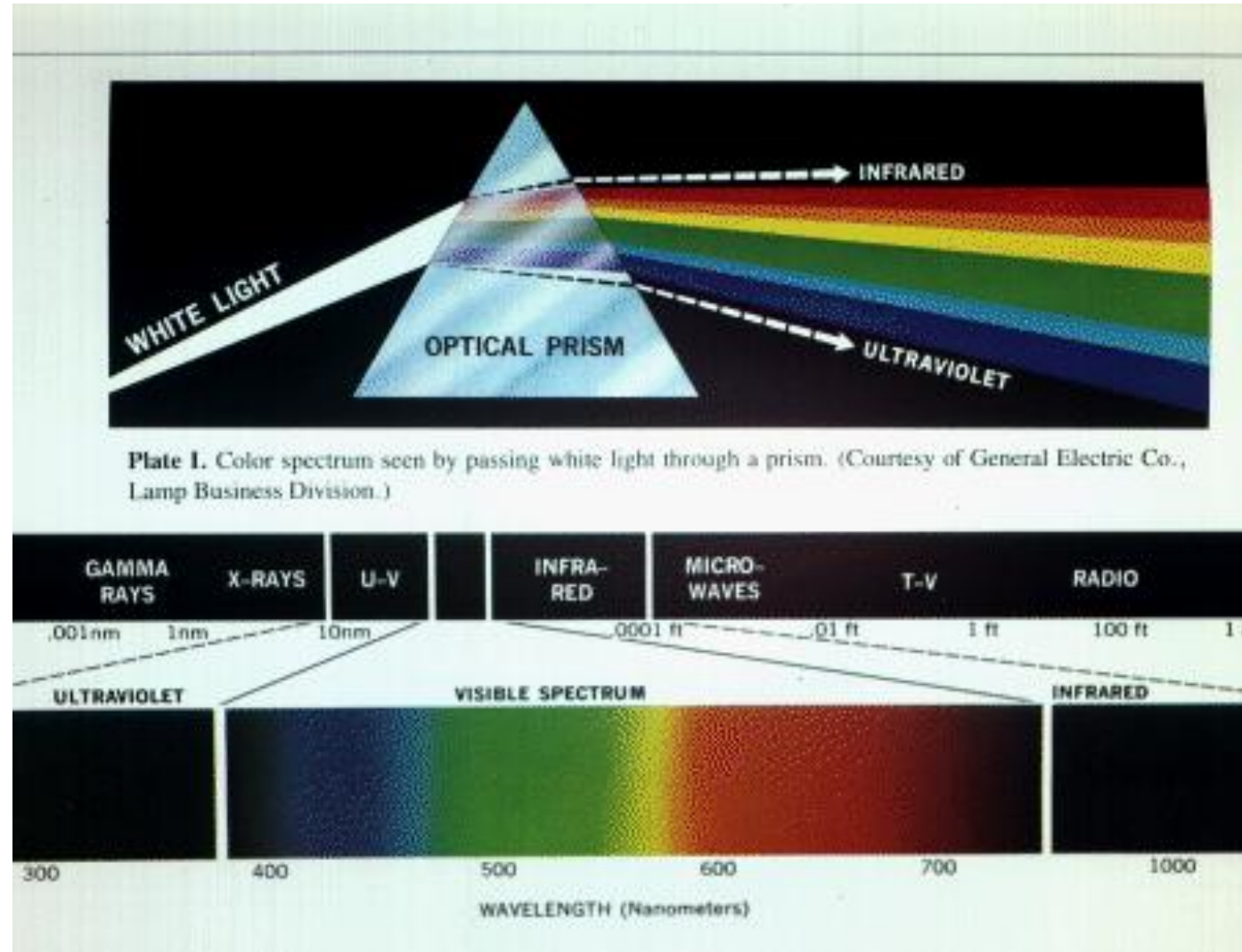
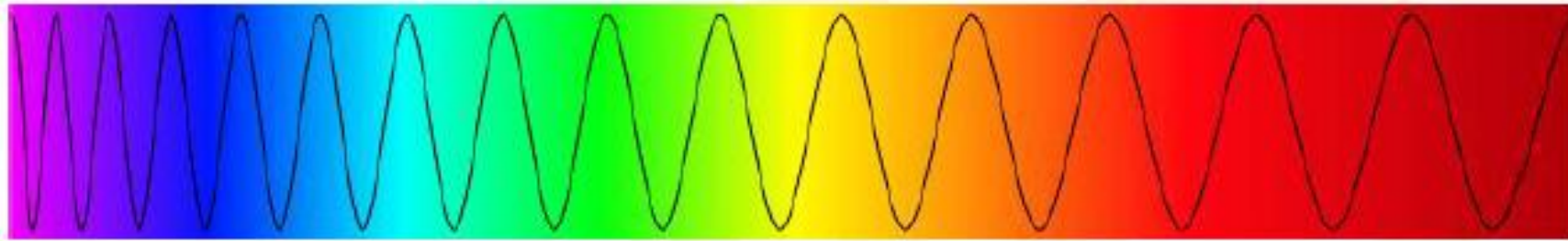


Image data

- Range of visible light: [380, 780] nm
- Blue: 435.8 nm, green: 546.1 nm, red: 700 nm

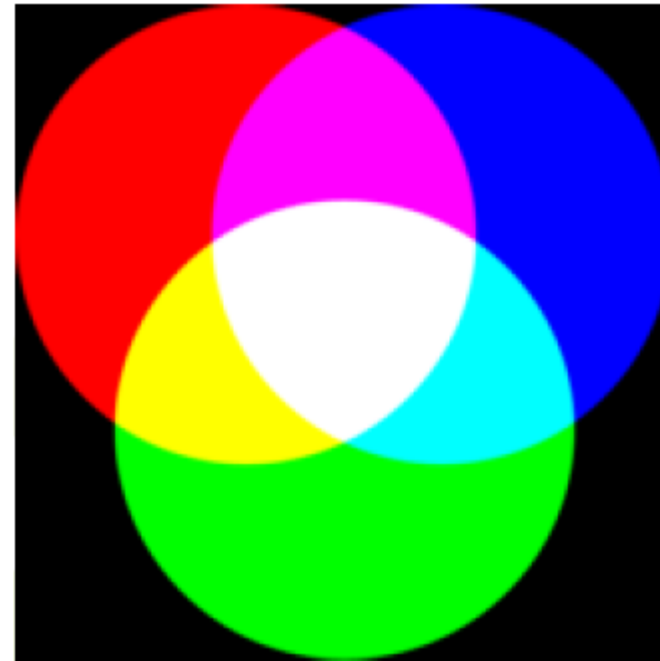


Colors and Light

- Human trichromatic vision uses three types of photoreceptor cone cells (S, M, and L) sensitive to short, medium, and long wavelengths.
- The RGB color model exploits this biological reality by additively mixing red, green, and blue light to stimulate those same overlapping cone pathways, tricking the brain into perceiving a full spectrum of colors
- Additive color (light): RGB — combine toward white
- Subtractive color (print): CMYK — combine toward black
- Perception $\neq$ physics: same RGB can look different under different illumination

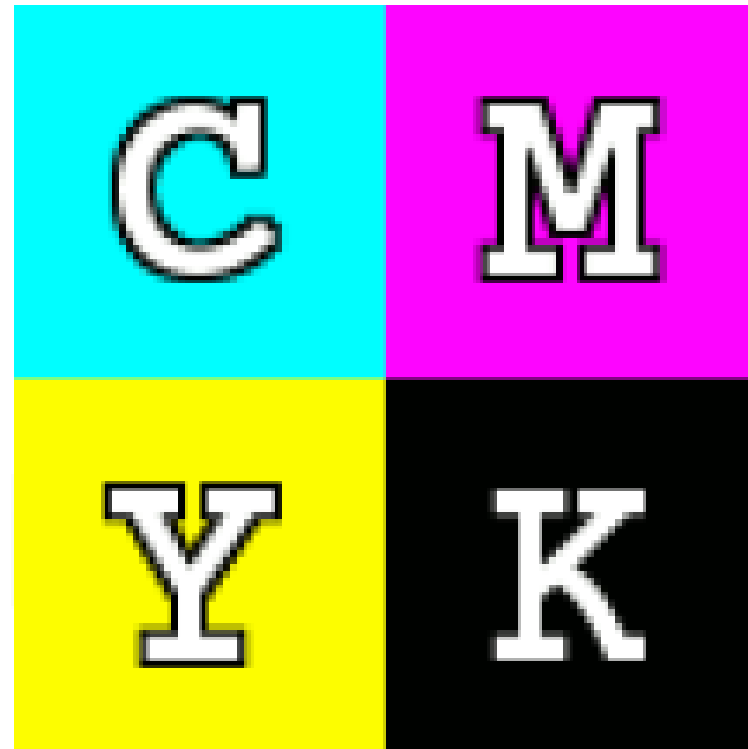
Color space

- RGB space
 - $(0,0,0)$: black
 - $(255,255,255)$: white
 - $(255,0,0)$: Red
 - $(0,255,0)$: Green
 - $(0,0,255)$: Blue
 - $(255,255,0)$: Yellow
 - $(0,255,255)$: cyan
 - $(255,0,255)$: magenta



Color space

- CMYK space: Cyan, Magenta, Yellow, Black



Color space

- CIE space (Commission International de l'Eclairage – Standardization Commission on Illumination): linear transformation from RGB space.
- HSV space (Hue, Saturation, Value): nonlinear transformation from RGB space.

Color Spaces — Comparison

- RGB: device-oriented; (255,255,255) white, (255,0,0) red (slide 7)
- CMYK: Cyan, Magenta, Yellow, Black — printing (slide 8)
- CIE: device-independent; linear transform from RGB (slide 9)
- HSV: Hue, Saturation, Value — intuitive for segmentation and CBIR
- MMDB: store raw format + normalized feature vectors in chosen space
- Conversion loss: gamut clipping when mapping between spaces

Image data

Describe the content of following images:



Image data

Characteristics of identification attributes:

- Assessment of differences among images
- Sometimes difficult to articulate in natural language
- May return multiple similar retrieval results
- Can generate complex queries by combining multiple attributes

Image retrieval

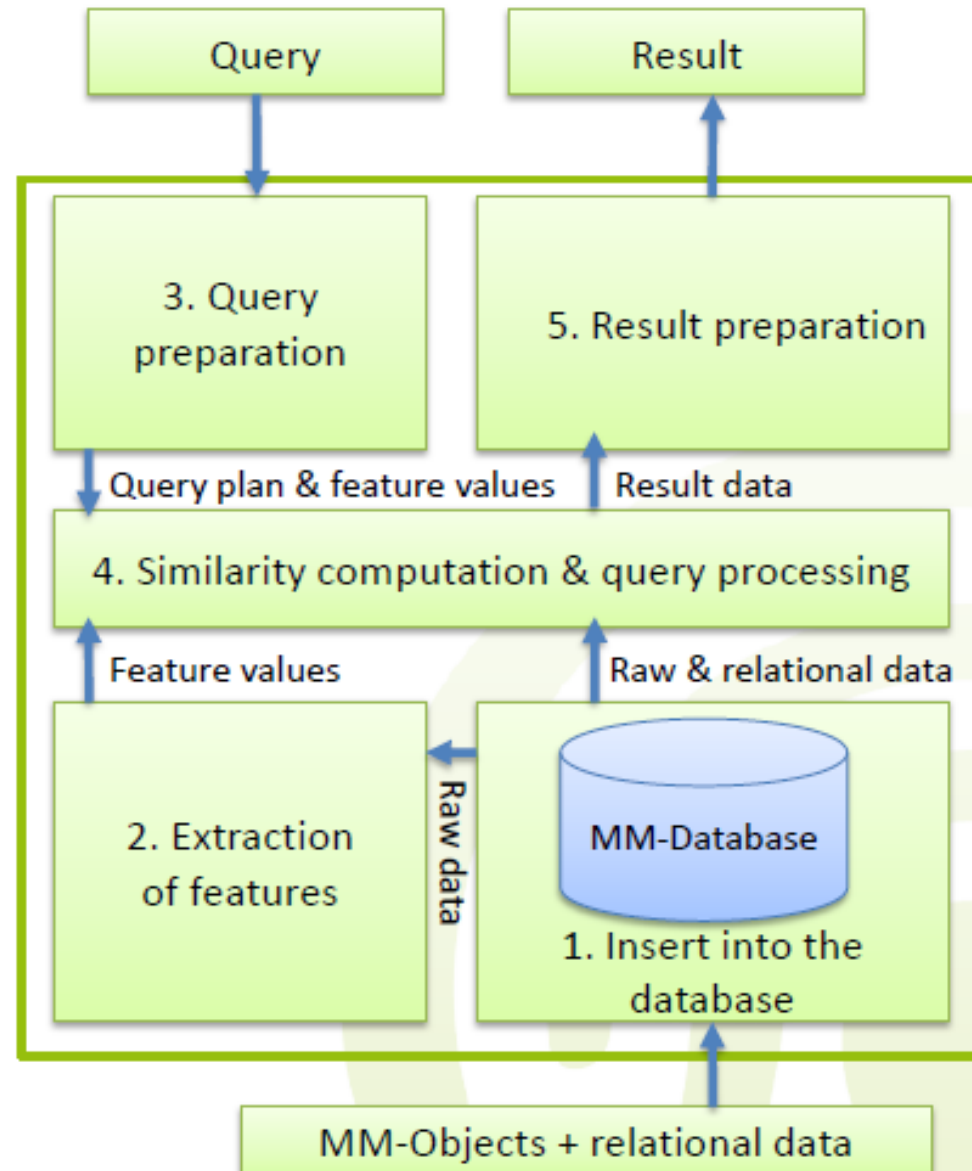


Image Retrieval — Pipeline Overview

- Query: image, sketch, or color/shape example
- Feature extraction: color, texture, shape, edges
- Similarity match in feature space → ranked results
- Challenges: semantic gap — orange frog vs green frog
- Combining low-level features improves accuracy but not full semantics

Image data

- Example of low-level attributes:
 - Color: object color, background color,...
 - Shape: size, shape of objects,...
 - Color distribution: contrast, color cover,...



Image data

Assumption: If two images have same colors, their contents are the same.

- Much of an image's semantic content may be lost if only low-level attributes are used for description,
- For example, describing a sunset image solely in terms of colors (e.g., orange and yellow) fails to capture its higher-level semantic meaning



The Semantic Gap in Image Retrieval

- Low-level features (color, texture) $\neq$ human concepts (sunset, wedding, tumor)
- Sunset described only as 'orange and yellow' loses meaning
- Color-only match can confuse similar palettes, different objects
- High-level attributes need models, tags, or deep features
- MMDB stores both: feature vectors + user/metadata annotations
- Relevance feedback helps bridge gap iteratively

Color comparison

- The comparison results may be satisfactory
 - for example, when an image of a green frog is compared with an image of a sunset
- However, the comparison results may be less reliable.
 - Example: an image of an orange-colored frog is compared with an image of a green frog.
 - In such cases, it may even appear similar to an image depicting a completely different type of content.



Image data

- Combining multiple low-level attributes can sometimes yield accurate retrieval results.
- However, a single visual frame may convey multiple semantic interpretations



Image data

- High-level attributes
 - Fourier transforms (DFT, FFT)
 - Treating the entire image as a signal sequence
 - Transforming from the spatial domain to the frequency domain (representing images in terms of intensity oscillations)
 - Preserving essential information without significant loss
 - Enabling the representation of information that is difficult to express verbally
 - Wavelet transforms (WT)
 - Discrete Cosine Transform (DCT)
 - ...

Transform-Domain Image Features

- DFT/FFT: spatial $\rightarrow$ frequency; intensity oscillations across image
- DCT: basis for JPEG; energy compaction in low-frequency coefficients
- Wavelet (WT): multi-resolution; used in JPEG 2000
- Capture patterns hard to describe in words (periodic texture, edges)
- Stored as feature vectors for similarity search
- Trade-off: transform cost at ingest vs faster retrieval

Audio data

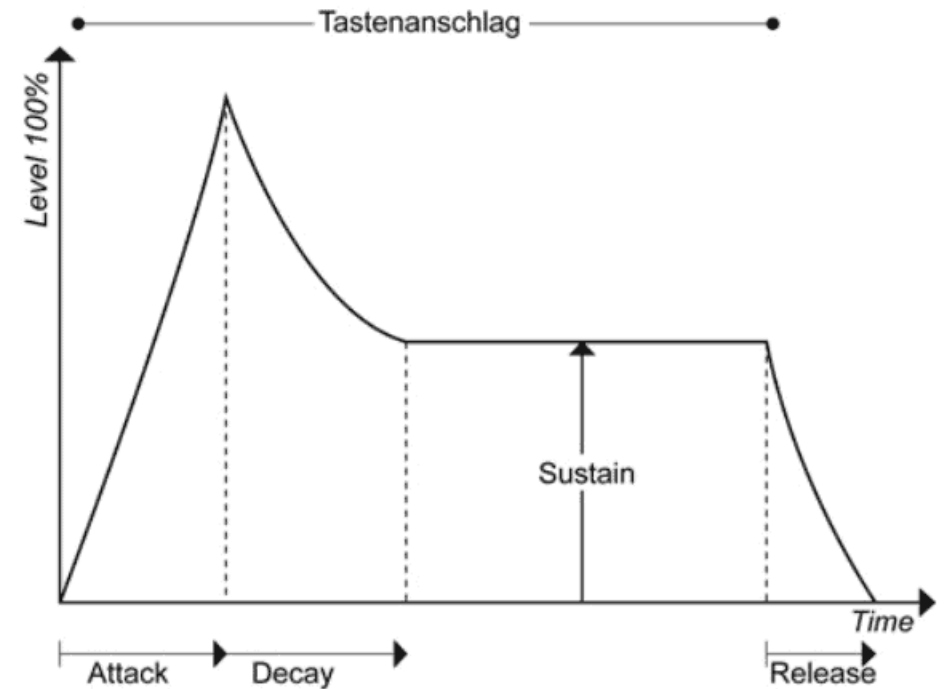
- What type of signal is sound?
- There are three main categories of audio data :
 - Music
 - Speech
 - Noise ...



Audio data

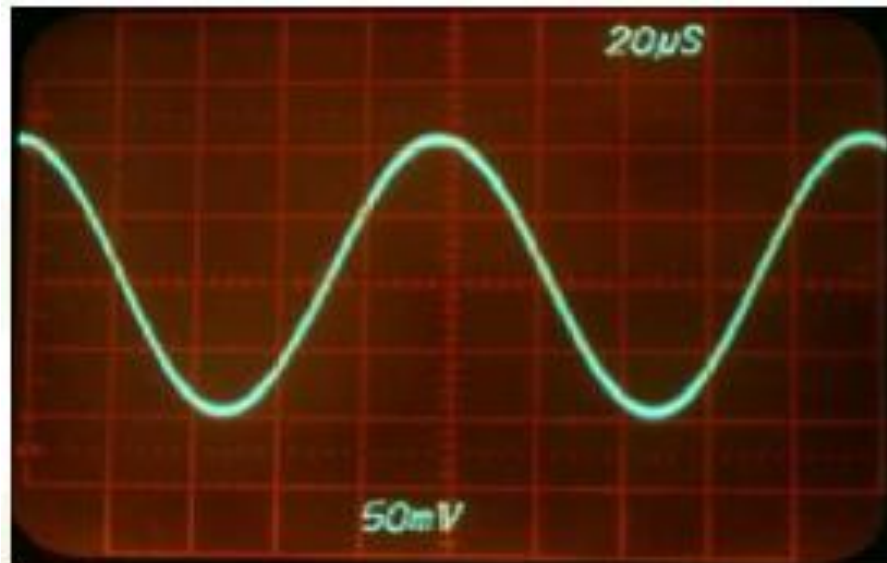
Sound generation

- An oscillating source
- An amplification mechanism
- A representation of sound intensity over time, commonly described by the Attack–Decay–Sustain–Release (ADSR) envelope



Audio data

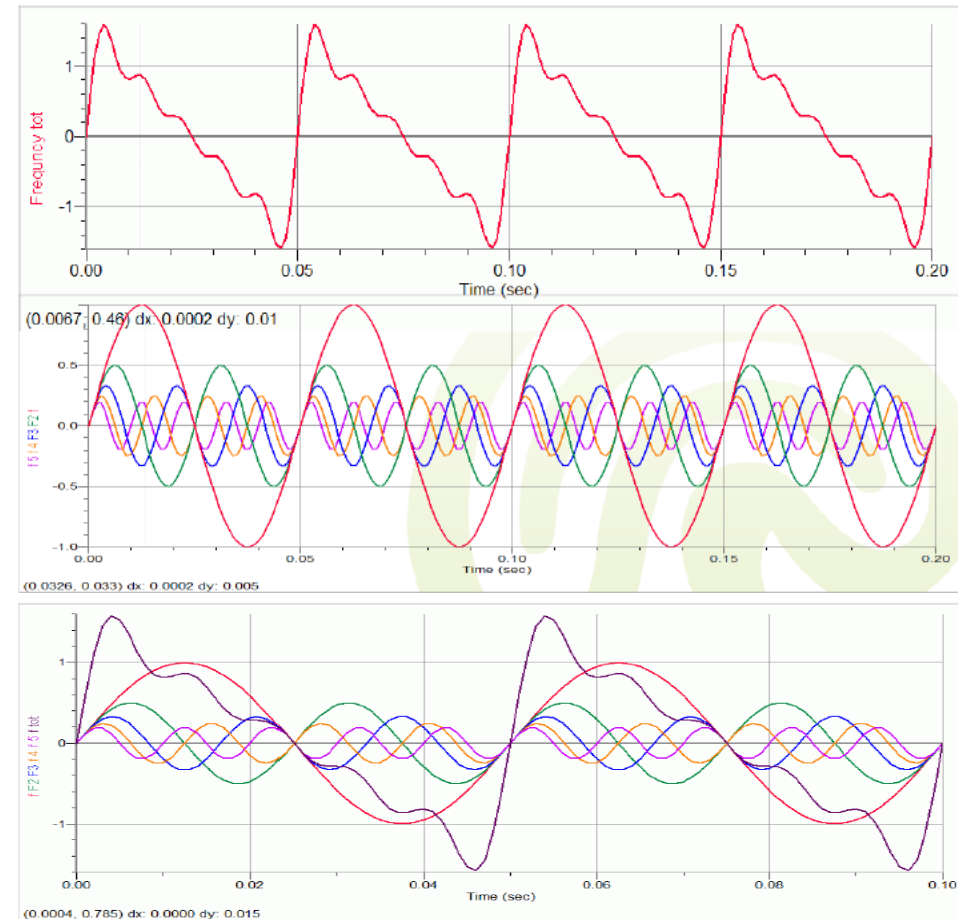
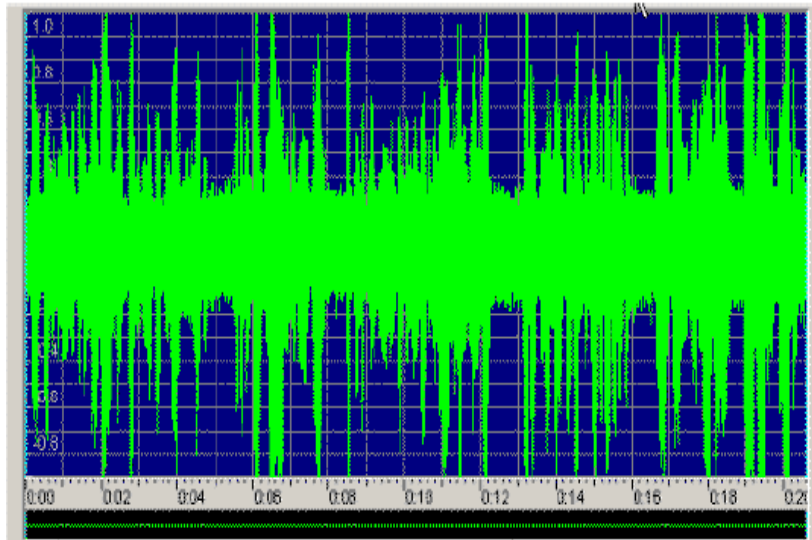
- The human brain can perceive only two fundamental attributes of sound:
 - **pitch**, which corresponds to the frequency of the sound wave (ranging from approximately 20 Hz to 20 kHz),
 - **loudness**, which is determined by the amplitude of the sound wave



Audio data

An audio signal is a composite of multiple sources, including:

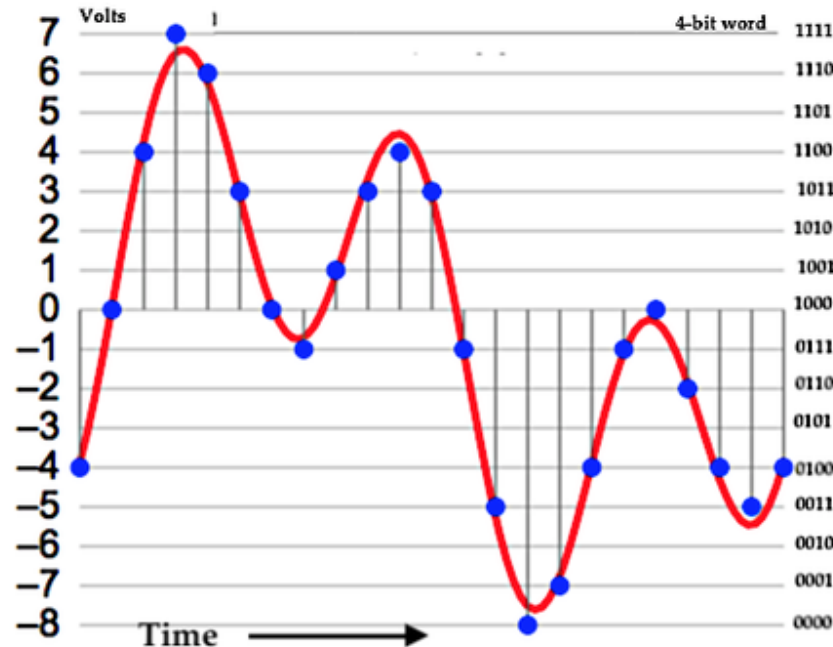
- information-bearing components,
- interfering noise,
- cancelling (destructive) noise



Audio data

Digitization of audio signals: converting a signal from a continuous form to a discrete representation. It consists of two main processes:

- **sampling**, defined by the sampling frequency (e.g., 44,100 Hz for audio CDs and 8,000 Hz for telephone systems)
- **quantization**, which is determined by the resolution (e.g., 16 bits).



CD-Quality Audio Bit-rate

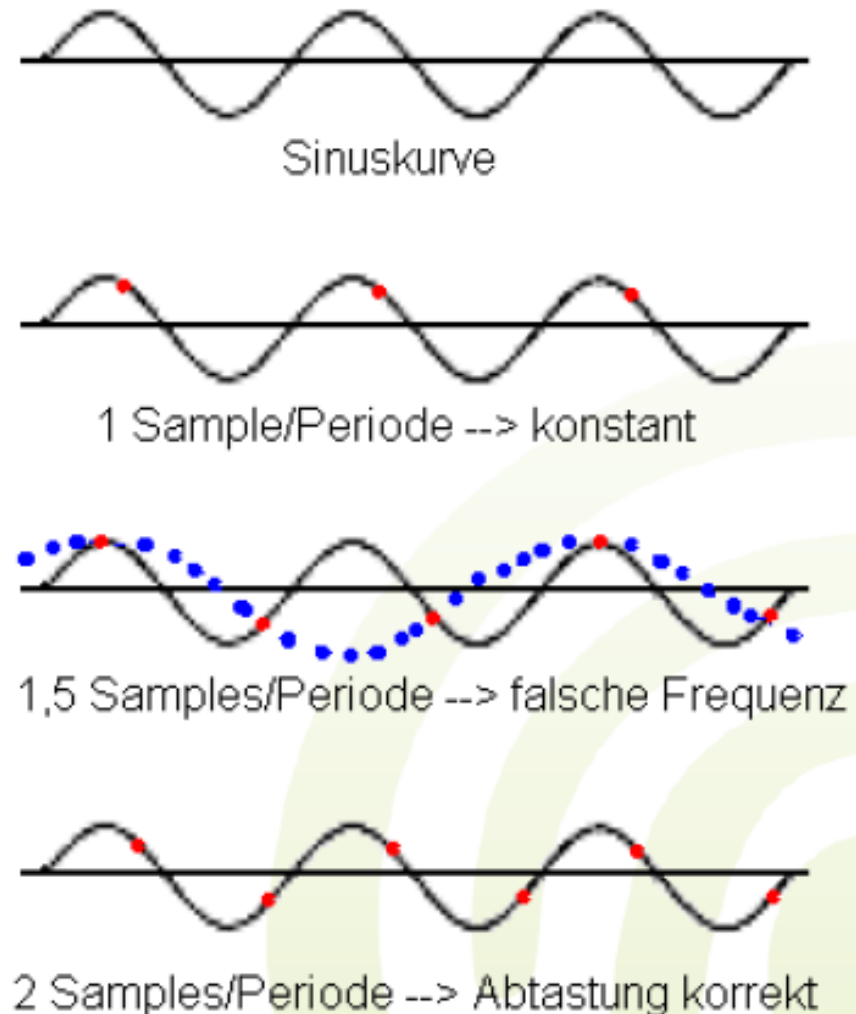
- Stereo audio, 44,100 Hz sampling, 16-bit quantization per sample per channel.
 - a) Bits per sample (both channels)?
 - b) Minimum data rate in bps? Convert to Mbps.
 - c) One hour of uncompressed stereo — size in MB?
 - d) Why is telephone audio (8 kHz, 8-bit mono) much smaller?

Nyquist Sampling Theorem

- Nyquist (1928): sample rate $\geq 2 \times$ highest signal frequency (slide 23)
- CD 44.1 kHz covers human hearing ~20 Hz–20 kHz
- Undersampling $\rightarrow$ aliasing: high frequencies fold into audible range
- Telephone 8 kHz $\rightarrow$ captures up to 4 kHz — sufficient for speech intelligibility
- MMDB: store sample rate + bit depth as mandatory audio metadata
- Re-encoding at lower rate is lossy — keep master archive at full quality

Audio data

- **Sampling theory** (Nyquist, 1928) states that the sampling frequency must be at least **twice the highest frequency component of the signal** in order to accurately reconstruct the original signal.



Audio data

Signal compression:

- Uncompressed signals: AIFF (*.aif), Wave (*.wav), IRCAM (*.snd), AU (*.au),...
- Lossy compression: Modified Discrete Cosine Transform (MDCT), wavelets,...
 - Encoding: transformation of the signal into the frequency domain
 - Decoding: reconstruction of the signal from its encoded representation
 - Information reduction:
 - removing high- or low-frequency components
 - reducing the precision of information in the retained frequency bands
 - exploiting psychoacoustic effects and other auditory models to discard perceptually less relevant information.

Audio data

Information contained in audio data

- Audio data
 - Music, CD...
 - Audio effects,...
- Audio documents
 - Speeches
 - Recorded conversational audio.

Audio data

Applications

- Audio signal recognition (querying);
- Classification and retrieval of similar audio signals;
- Audio signal synchronization

Audio data

Low level attributes:

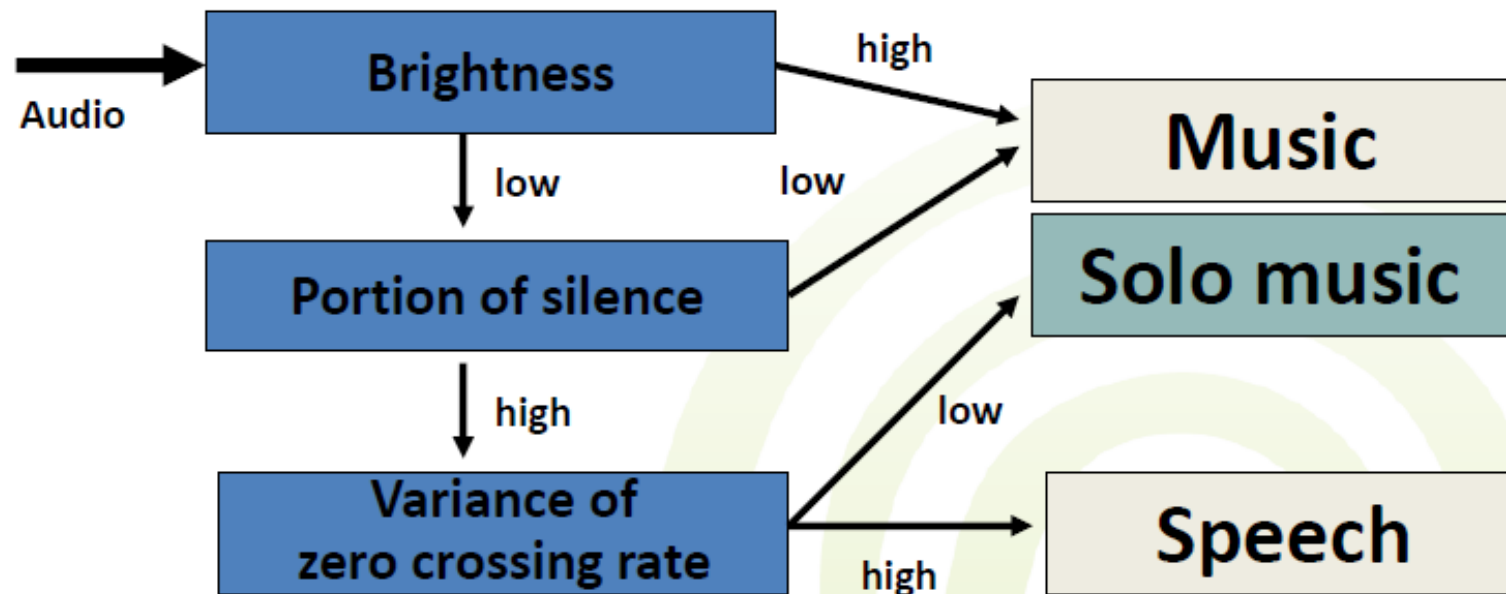
- Sound intensity;
- Frequency distribution of audio signals;
- Pitch;
- Clarity;
- Average energy
- ...

Low level attributes can be obtained from

- Time domain signals
- Frequency domain signals

Audio data

- Example of classifying audio data based on low level attributes



Text data

- **Plain text:** a sequence of characters..
 - Character encoding schemes : ASCII code (8 bits)
 - Example: How much storage is required to store **300 pages of text**, with **3,000 characters per page**?
- **Structured text** includes formats such as:
 - HyperText Markup Language (HTML)
 - Standard General Markup Language (SGML)
 - LaTeX
 - Office Document Architecture (ODA)
 - Portable Document Format (PDF)

Text Data — Storage and Encoding

- Plain text: character sequence; ASCII 8 bits per character
- 300 pages × 3,000 chars = 900,000 characters → 900 KB (ASCII)
- Unicode UTF-8: 1–4 bytes per character — plan for Vietnamese/emoji
- Structured text: HTML, SGML, LaTeX, ODA, PDF — markup + content
- MMDB: full-text index on extracted plain text
- PDF stores compressed streams — extract text for search, keep BLOB for display

Text Storage Calculation

- 300 pages, 3,000 characters per page, ASCII encoding:
 - a) Total characters?
 - b) Storage in bytes? KB? MB?
 - c) Same document UTF-8 (avg 2 bytes/char Vietnamese) — new size?
 - d) HTML version 40% markup overhead — estimated total?

ASCII codebook

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

ASCII codebook

128	Ç	144	É	160	á	176	☐	192	Ł	208	⌌	224	α	240	≡
129	ü	145	æ	161	í	177	☐	193	Ł	209	⌌	225	β	241	±
130	é	146	Æ	162	ó	178	☐	194	Ł	210	⌌	226	Γ	242	≥
131	â	147	ô	163	ú	179		195	Ł	211	⌌	227	π	243	≤
132	ä	148	ö	164	ñ	180	Ł	196	Ł	212	⌌	228	Σ	244	∫
133	à	149	ò	165	Ñ	181	Ł	197	Ł	213	⌌	229	σ	245	∫
134	â	150	û	166	²	182	⌌	198	Ł	214	⌌	230	μ	246	÷
135	ç	151	ù	167	°	183	⌌	199	Ł	215	⌌	231	τ	247	≈
136	ê	152	ÿ	168	¿	184	Ł	200	⌌	216	⌌	232	Φ	248	°
137	ë	153	Ö	169	┐	185	⌌	201	⌌	217	┐	233	⊙	249	.
138	è	154	Û	170	┐	186	⌌	202	⌌	218	┐	234	Ω	250	.
139	ì	155	◊	171	½	187	Ł	203	⌌	219	■	235	δ	251	√
140	î	156	£	172	¼	188	⌌	204	⌌	220	■	236	∞	252	π
141	ï	157	¥	173	¡	189	⌌	205	=	221	■	237	φ	253	²
142	Ä	158	£	174	«	190	┐	206	⌌	222	■	238	ε	254	■
143	Å	159	ƒ	175	»	191	┐	207	⌌	223	■	239	∩	255	

Source: www.LookupTables.com

Video data

- **Requirements for a video database:**
 - storage capability
 - data accessibility
 - video recovery and playback support
- **Common types of video databases**
 - Blobs (binary large objects)
 - metadata-based retrieval systems
 - Video segmentation



Video Database Architecture

- Requirements: storage, accessibility, recovery, playback
- BLOB store for compressed video files; DB for metadata + indexes
- Segmentation: shots, scenes, keyframes
- Multi-stream sync: video + audio + subtitles
- Continuous media: stream from disk/network, not random page fetch
- Tools: DB2 extenders, QBIC integration — unified query interface

Metadata vs Video BLOB — Storage Split

- DB stores: frame rate, codec, keyframe count, duration, title
- File/object store holds: MPEG/H.264 bitstream
- Retrieval by metadata first; decode video only for matched candidates
- Content-based: keyframe features link to indexes
- Avoid storing raw frames in relational rows — reference by URI/OID
- Panofsky metadata layers apply to video documents

Video data

- Supporting tools
 - **IBM AIV Extenders for IBM DB2 UDB**, which enable the integration of video data with commercial data within a single query.
 - *Example:* retrieving all video segments related to the same topic and sorting them by recording time.
 - **Integration of QBIC (Query By Image Content) configurations** into commercial database systems.

Video data

- How is video database built?
 - Video attributes are inserted into and retrieved from the database
 - the database itself stores and manages only the **metadata and descriptive attributes** of the video: frame rate, image compression format, and the number of key frames,...
 - **Video retrieval** is performed based on the stored attributes, including identifiers such as name and index number, content descriptors, or format-related properties.
 - Include tools supporting video capture

Video data

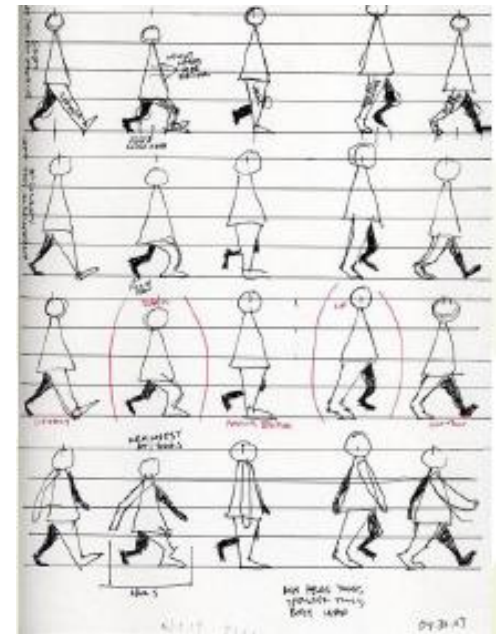
- Issues to be addressed :
 - continuous transmission media
 - multiple communication channels
 - transmission of image sequences
 - simultaneous transmission of audio
 - simultaneous transmission of text



Video data

Problems to be solved

- Information Organization
- Video transfers structured information both in space and time
- Video is a form of document
- The video consists of many different structured sections



Video data

- Video retrieval is a compilation of all the retrieval tools:
 - Image
 - Audio
 - Text

Video data

- Attributes used for video retrieval
 - object recognition based on segmentation and subtitles
 - specific contextual information (e.g., audience, environment, spatial setting)
 - direction and speed of object motion
 - camera movement and changes in viewpoint or angle
 - spatial and temporal relationships
 - integration of audio data
 - speech characteristics and semantic content
 - ...



Summary — Multimedia Data Types

- Image: 2D matrix; color spaces; semantic gap between pixels and meaning
- Audio: time-series; sample rate + quantization; Nyquist; MFCC features
- Text: character encoding; plain vs structured; efficient but needs IR index
- Video: integrates image + audio + text; metadata/BLOB split essential
- All types: low-level features for CBIR/CBAR; high-level metadata for semantics